

High-frequency data and limit order books



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CentraleSupélec

U. Paris-Saclay CentraleSupélec cursus Ingénieur 3A Mathématiques et Data Science
IP Paris ENSAE M2 Statistics, Finance and Actuarial Science
U. Paris-Saclay Evry M2 Quantative Finance

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Realized variance I

- Consider a (driftless) stochastic volatility model for a (log-)price p_t , $t \in [0, T]$:

$$p(t) = \int_0^t \sigma_s dB_s.$$

- The variance integrated over the time interval $[0, T]$ is

$$IV = [p]_T = \int_0^T \sigma_s^2 ds.$$

- For a given sampling period τ , we naturally define the realized variance estimator:

$$RV(\tau) = \sum_{i=1}^{\lfloor T/\tau \rfloor} (p(i\tau) - p((i-1)\tau))^2.$$

Realized variance II

- ▶ $RV(\tau)$ is a consistent estimator of IV :

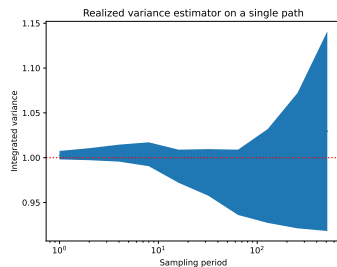
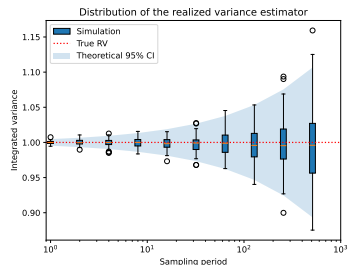
$$RV(\tau) \xrightarrow[\tau \rightarrow 0]{} IV \text{ in probability.}$$

- ▶ The estimator $RV(\tau)$ is asymptotically Gaussian [3]:

$$\frac{RV(\tau) - IV}{\sqrt{\frac{2}{3} \sum_{i=1}^{\lfloor T/\tau \rfloor} (p(i\tau) - p((i-1)\tau))^4}} \xrightarrow[\tau \rightarrow 0]{\mathcal{L}} \mathcal{N}(0, 1).$$

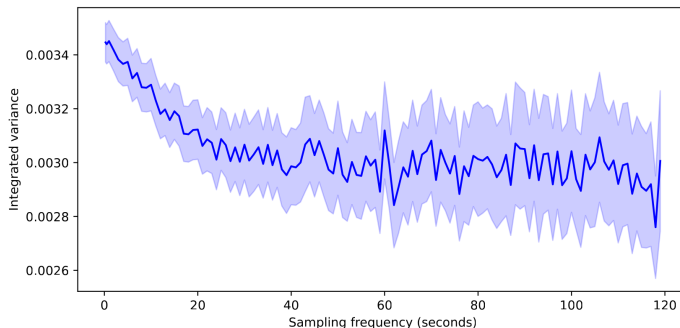
This gives us a measure of the estimation error for the realized variance (volatility).

- ▶ *Right column:* illustration on $\frac{1}{\sqrt{T}}B_t$ (scaled Brownian motion) simulated on $[0, T]$, $T = 3.5E5$.



Volatility signature plot

- ▶ However on financial markets $RV(\tau)$ is empirically increasing when τ decreases. This is known as the **volatility signature plot**. Example on a French stock (2017 data):



- ▶ RV estimation on high-frequency financial data in contradiction with the Brownian paradigm: **“microstructure noise”**.

A Hawkes model for the signature plot I

- ▶ [1] proposes a model of Hawkes-driven upwards and downwards price jumps.
- ▶ Price is defined as $p(t) = N^1(t) - N^2(t)$, with $N^i, i \in \{1, 2\}$, Hawkes processes such that

$$\begin{cases} \lambda^1(t) = \lambda_0 + \int_{-\infty}^t \alpha e^{-\beta(t-s)} dN_s^2, \\ \lambda^2(t) = \lambda_0 + \int_{-\infty}^t \alpha e^{-\beta(t-s)} dN_s^1. \end{cases}$$

- ▶ No self-excitation of upward (resp. downward) jumps on following upward (resp. downward) jumps
- ▶ Only cross-excitation terms are kept, enforcing the mean-reversion empirically observed on the price p
- ▶ Cross-excitation is set to be symmetric.

A Hawkes model for the signature plot II

- ▶ The **volatility signature plot** plots the (normalized) realized variance as a function of the sampling period:

$$\hat{C}(\tau) = \frac{1}{T} \sum_i^{T/\tau} (\hat{p}(i\tau) - \hat{p}((i-1)\tau))^2.$$

- ▶ **Mean signature plot** in a stationary regime:

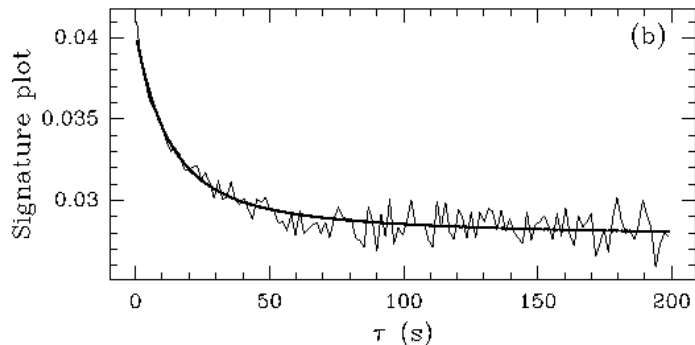
$$C(\tau) = \mathbf{E}[\hat{C}(\tau)] = \frac{1}{\tau} \mathbf{E} [p(\tau)^2]$$

- ▶ In the case of the above mean-reversion Hawkes model:

$$C(\tau) = \Lambda \left(\kappa^2 + (1 - \kappa^2) \frac{1 - e^{-\gamma\tau}}{\gamma\tau} \right),$$

with $\Lambda = \frac{2\lambda_0}{1-\alpha/\beta}$, $\kappa = \frac{1}{1+\alpha/\beta}$, and $\gamma = \alpha + \beta$.

A Hawkes model for the signature plot III



Source: [1]

- ▶ Empirical data (Bund futures contracts 2009). Data: last bid price only (no bid-ask bounce).
- ▶ Signature plot reproduced by a Hawkes price model (Warning: result is not a MLE fit, but fit of the signature plot)

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Correlation estimation with high-frequency data

- ▶ High-frequency financial data is non-synchronous: trades on stock 1 and trades on stock 2 do not occur at the same time.
- ▶ Estimation can be done via synchronous sampling, i.e. taking the last observed trade price at each sampling time, at the risk of introducing biases.
- ▶ If two continuous martingales P_1 and P_2 are observed on $[0, T]$ at asynchronous random times, the standard realized covariance estimator using the last observation is a biased estimator of $\mathbf{E}[\langle P_1, P_2 \rangle_T]$ (See [12]).
- ▶ A few estimators have been proposed to directly use asynchronous data.

The Hayashi-Yoshida estimator I

- ▶ Proposed in [12].
- ▶ Consider two processes ($k = 1, 2$):

$$dP_k(t) = \mu_k(t)dt + \sigma_k(t)dW_k(t)$$

with $(W_k(t))_t$ Brownian motions and $d\langle W_1, W_2 \rangle_t = \rho(t)dt$, for some deterministic functions ρ , σ_1 , and σ_2 .

- ▶ Assume that we observe the process k on $[0, T]$ with at random times $T^{k,i}$. Let $I^i = (T^{1,i}, T^{1,i+1})$ and $J^i = (T^{2,i}, T^{2,i+1})$. Let n be the total number of observations.
- ▶ Let $\Delta P_1(I^i)$ be the increment of P on I^i (and define similarly $\Delta P_2(J^i)$). The Hayashi-Yoshida covariance estimator is

$$U_n = \sum_{i,j} \Delta P_1(I^i) \Delta P_2(J^j) \mathbf{1}_{I^i \cap J^j \neq \emptyset}.$$

The Hayashi-Yoshida estimator II

- ▶ The estimator is quite easy to implement.
- ▶ The estimator is consistent [12]:

$$U_n \rightarrow \int_0^T \sigma_1(t)\sigma_2(t)\rho(t)dt \text{ in probability when } n \rightarrow +\infty.$$

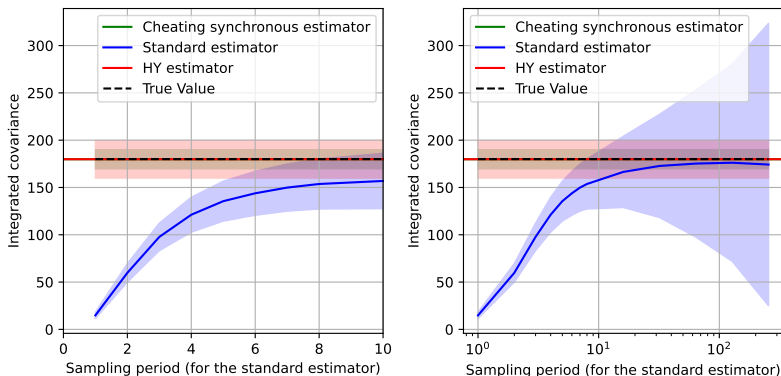
- ▶ The estimator is asymptotically Gaussian [11].
- ▶ In the case of constant ρ , σ_1 and σ_2 , the correlation estimator is also consistent:

$$\frac{\sum_{i,j} \Delta P_1(I^i) \Delta P_2(J^j) \mathbf{1}_{I^i \cap J^j \neq \emptyset}}{\sqrt{\sum_i \Delta P_1(I^i)^2} \sqrt{\sum_j \Delta P_2(J^j)^2}} \rightarrow \rho \text{ in probability when } n \rightarrow +\infty.$$

(but careful, no asymptotic theory and may be numerically unstable)

The Hayashi-Yoshida estimator III

- An illustration on Brownian motions with Poisson sampling:

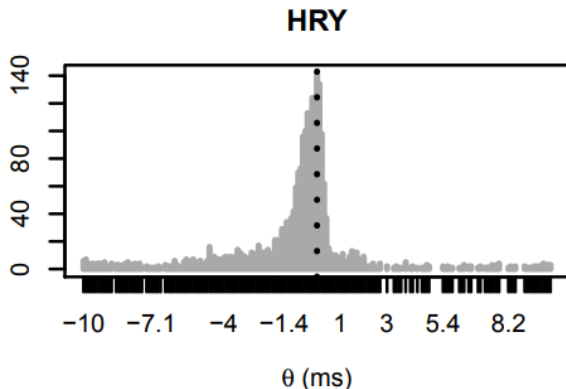


- On a similar subject, see also the Fourier (Malliavin-Mancino) estimator [16].

Lead-lag estimation using covariance estimators

- Covariance can be used to estimate a lead-lag relationship between two stocks.
- Let P_1 and P_2 be two timeseries of high-frequency prices, and $U_n(P_1, P_2)$ be the Hayashi-Yoshida covariance estimator.
- For $\ell \in \mathbb{R}$, let $P_2^{(\ell)}$ be the timeseries P_2 shifted by a lag ℓ .
- Then $\hat{\ell} = \operatorname{argmax}_{\ell} U_n(P_1, P_2^{(\ell)})$ is an estimator of the lead-lag relationship between P_1 and P_2 (see e.g., [13]).

Source: [10]

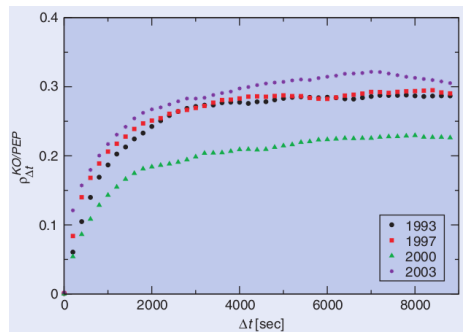


The Epps effect

- [5]: correlation measured on financial assets decreases when the sampling frequency increases (1979). Still observed on modern markets (e.g., [22], right panel below).

1. Correlations of Changes in Log Price for Four Stocks During Intervals of 10 Minutes to Three Days

Interval	Pairs of Stocks					
	AMC-Chrysler	AMC-Ford	AMC-GM	Chrysler-Ford	Chrysler-GM	Ford-GM
10 minutes	.001	.009	-.009	-.014	.007	.055
20 minutes	.009	.018	.011	.017	.026	.118
40 minutes	.006	.012	.014	.041	.040	.197
One hour	-.043	.057	.064	.023	.065	.294
Two hours	.029	.060	.094	.112	.129	.383
Three hours	.031	.158	.111	.361	.518	.519
One day	-.067	.170	.078	.342	.442	.571
Two days	-.020	.223	.186	.336	.449	.572
Three days	-.098	.203	.100	.334	.542	.645



- Asynchronicity, lead-lag, ...

The Epps effect in a Hawkes model I

- [1] investigates the Epps effect using a bivariate version of the previous model:

$$\begin{cases} p_1(t) &= N^1(t) - N^2(t), \\ p_2(t) &= N^3(t) - N^4(t), \end{cases}$$

in which $\mathbf{N} = (N^i)_{i=1,\dots,4}$ is a 4D-Hawkes process with intensity:

$$\lambda(t) = \lambda_0 + \int_0^t \begin{pmatrix} 0 & \phi^{12} & \phi^{13} & 0 \\ \phi^{12} & 0 & 0 & \phi^{13} \\ \phi^{31} & 0 & 0 & \phi^{34} \\ 0 & \phi^{31} & \phi^{34} & 0 \end{pmatrix} (t-s) d\mathbf{N}_s, \quad (1)$$

where $\phi^{ij}(t-s) = \alpha^{ij} e^{-\beta^{ij}(t-s)}$.

The Epps effect in a Hawkes model II

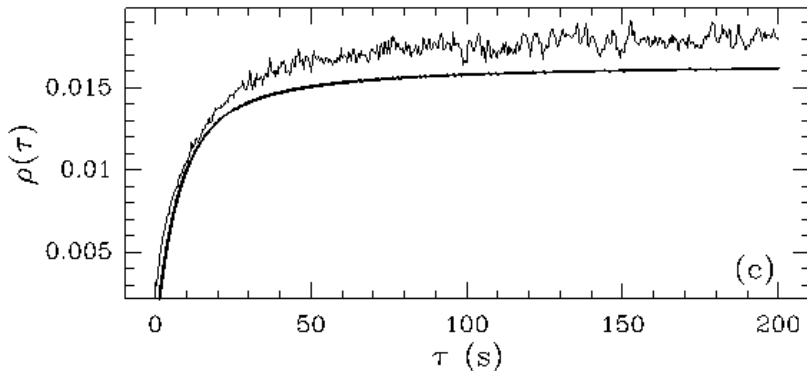
- ▶ Self-exciting terms are ruled out, $\phi^{ii} = 0$ for all i .
- ▶ **Symmetric mean-reversion** for each price process p_1 and p_2 ($\phi^{12} = \phi^{21}$, $\phi^{34} = \phi^{43}$).
- ▶ Only **positive** influence between prices p_1 and p_2 ($\phi^{14} = \phi^{23} = \phi^{32} = \phi^{41} = 0$).
- ▶ One price may have on stronger influence on the other, but this influence is **symmetric w.r.t. the direction of the move** ($\phi^{24} = \phi^{13}$, $\phi^{42} = \phi^{31}$).
- ▶ **Explicit form of the correlation coefficient available**

$$\rho(\tau) = \text{Corr}(p_1(t + \tau) - p_1(t), p_2(t + \tau) - p_2(t)),$$

although quite cumbersome to write (see [1, Proposition 3.1]).

The Epps effect in a Hawkes model III

- Hawkes model in good agreement with the Epps effect. Bund/Bohl example:



Source: [1]

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Reminder: Branching representation of a Hawkes process

Proposition (Branching representation of a Hawkes process)

Let $T > 0$. Let $\mu : [0, \infty) \rightarrow \mathbb{R}_+$ a deterministic baseline function and $\nu : [0, \infty) \rightarrow \mathbb{R}_+$ a deterministic function. Consider the following procedure:

1. Generate a non-homogeneous Poisson process $\{t_i^{(0)}\}$ with intensity μ on $[0, T]$.
2. For each $t_i^{(0)}$, generate a non-homogeneous Poisson process $\{t_i^{(1)}\}$ with deterministic intensity $t \mapsto \nu(t - t_i^{(0)})$ on $[t_i^{(0)}, T]$.
3. Repeat operation 2. for each point of all the sets $\{t_i^{(1)}\}$, then for each point of all the sets $\{t_i^{(2)}\}$, etc. until there is no new point generated in $[0, T]$.

Then the ordered set of all the points generated in $[0, T]$ is a Hawkes process N on $[0, T]$ with intensity $\lambda(t) = \mu(t) + \int_0^t \nu(t-s) dN_s$.

Branching ratio, endogeneity

- ▶ Each (parent) point generates an average of $\int_0^{+\infty} \nu(t) dt = \|\nu\|_1$ (children) points. $\|\nu\|_1$ is called the **branching ratio** of the branching process.
- ▶ The average intensity is $\bar{\lambda} = \frac{\mu}{1 - \|\nu\|_1}$. The ratio of endogeneous events in a stationary sample (**endogeneity ratio** ρ) is thus

$$\rho = 1 - \frac{\mu}{\bar{\lambda}} = \|\nu\|_1.$$

- ▶ In other words, $\mathbf{E}[dN_t^{exo}] = \mu dt$ and $\mathbf{E}[dN_t^{endo}] = \bar{\lambda} \|\nu\|_1 dt$.
- ▶ In a financial context, endogenous vs. exogenous trading. Careful calibration needed. Are financial markets in a critical regime ? See eg. [8, 9, 7].

Causality (lead-lag)

- ▶ Let us consider a stationary D -dimensional Hawkes process with constant baseline intensities $\mu = (\mu^i)_{i=1,\dots,D}$ and kernel matrix $\Phi = (\Phi_{ij})_{i,j=1,\dots,D}$.
- ▶ Let N_t^{i0} be the counting process of events of type i generated exogenously. Then

$$\mathbf{E}[dN_t^{i0}] = \mu^i dt.$$

- ▶ Let N_t^{ij} be the counting process of events of type i with a direct ancestor of type j .

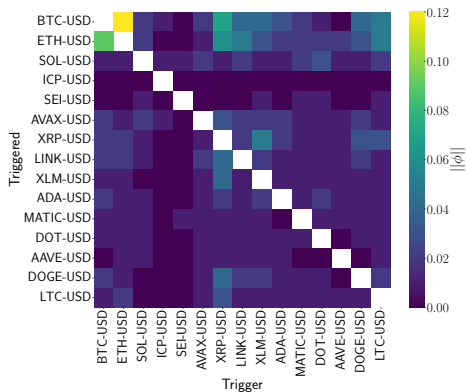
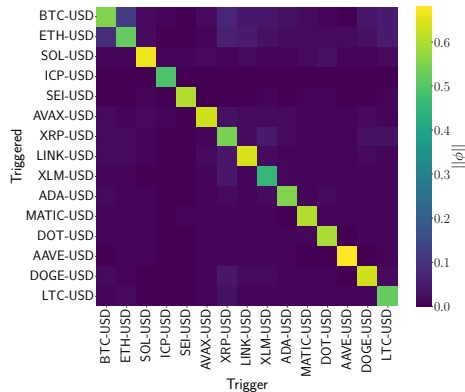
$$\mathbf{E}[dN_t^{ij}] = \|\Phi_{ij}\|_1 \Lambda_j dt.$$

where $\Lambda = (\mathbf{I} - \|\Phi\|_1)^{-1} \mu$ is the vector of average intensities.

- ▶ This provides a **causality** measure that can be used on financial markets.

An example of analysis on cryptocurrency pairs

- Endogeneity matrices on trade times of 15 cryptocurrency pairs on a centralized exchange.



Source : [6]

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A Hawkes model for aggressive orders I

- [14] is an early attempt at modeling many types of LOB events with a multidimensional Hawkes process. Events are divided into ten categories (*right*).

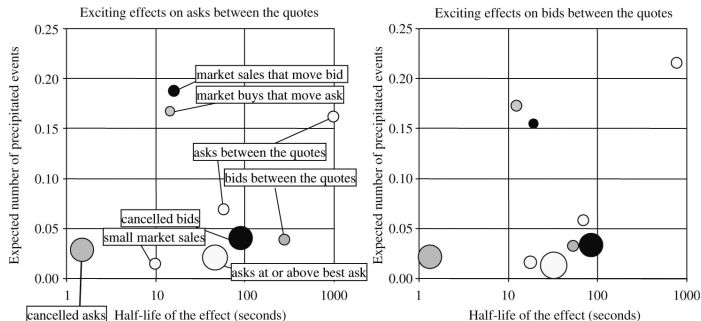
Type	Description	Aggressiveness
1	Market order that moves the ask	Yes
2	Market order that moves the bid	Yes
3	Limit order that moves the ask	Yes
4	Limit order that moves the bid	Yes
5	Market order that doesn't move the ask	No
6	Market order that doesn't move the bid	No
7	Limit order that doesn't move the ask	No
8	Limit order that doesn't move the bid	No
9	Cancellation at ask	No
10	Cancellation at bid	No

- A 4-dimensional Hawkes processes for aggressive orders is proposed ($m = 1, \dots, 4$):

$$\lambda^m(t) = \lambda_0(t) + \sum_{n=1}^{10} \int_0^t \alpha_{mn} e^{-\beta_{mn}(t-u)} dN_u^n.$$

A Hawkes model for aggressive orders II

► Estimates for aggressive limit orders

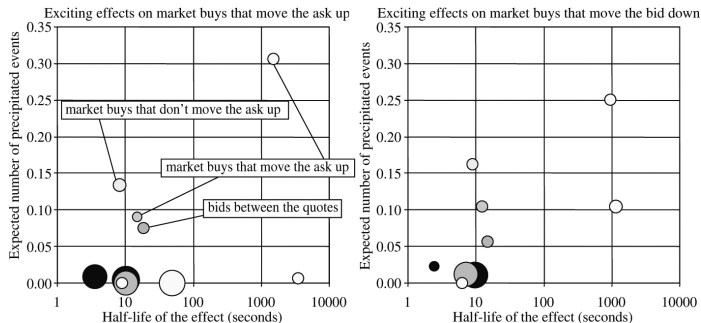


► Reported findings:

- aggressive limit orders are firstly influenced by aggressive market orders: this is an evidence of some **resiliency** or **market making** in the order book;
- aggressive limit orders are secondly influenced by aggressive limit orders.

A Hawkes model for aggressive orders III

► Estimates for aggressive market orders



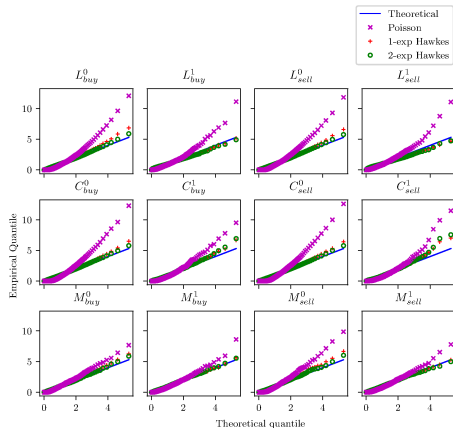
► Reported findings:

- aggressive market orders are firstly influenced by market orders (aggressive or not);
- aggressive market orders are secondly influenced by aggressive limit orders, which might be labeled a **rush to liquidity**.

A general Hawkes LOB model – 0,1 event description for the first limits

- ▶ [15] proposes a large Hawkes model with all events happening at the **best quotes**.
- ▶ Three types of orders (M,L,C), two aggressivities (0/non-aggressive,1/aggressive) and two sides (bid/ask) yield **12 event types**: $E = \{L_{buy}^0, L_{sell}^0, C_{buy}^0, C_{sell}^0, M_{buy}^0, M_{sell}^0, L_{buy}^1, L_{sell}^1, C_{buy}^1, C_{sell}^1, M_{buy}^1, M_{sell}^1\}$.
- ▶ 12-dimensional Hawkes process $N(t) = (N_{L_{buy}^0}(t), \dots, N_{M_{sell}^1}(t))$, with intensity process $(\lambda_{L_{buy}^0}(t), \dots, \lambda_{M_{sell}^1}(t))$.
- ▶ Variants tested: 1 or 2 exponential kernels:
 - ▶ $\phi_{ij}(t) = \alpha_{ij} \exp(-\beta_{ij}t)$
 - ▶ $\phi_{ij}(t) = \sum_{p=1}^2 \alpha_{ijp} \exp(-\beta_{ijp}t)$

A general Hawkes LOB model – Goodness-of-fit



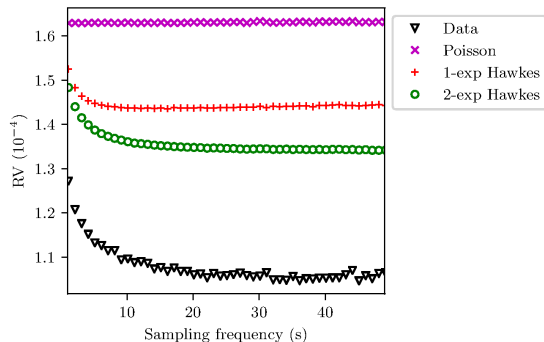
- ▶ Transformed durations should be **exponentially distributed** (see previous lecture)
- ▶ Q-Q plot goodness of fit tests of order book models
- ▶ Poisson model globally fails to capture the distributional properties of interval times.
- ▶ 1- and 2-exponential Hawkes models: similar performances, except for orders of type 0
- ▶ 1- and 2-exponential Hawkes models: close to one another, but not very close to data

A general Hawkes LOB model – Mid-price approximation

- ▶ LOB is not modelled *per se* (no tracking of limits)
- ▶ **Simplifying assumption** : all bid and ask price movements are equal to one tick (ticksize $\eta > 0$)
- ▶ **OK for large tick stocks** ; in the paper sample: 1.08 ticks per jump in average.
- ▶ (A proxy for the) **mid-price dynamics** is obtained as a by-product of event arrivals:

$$P(t) = P(0) + \left(\sum_{e \in E_{up}} N_e(t) - \sum_{e' \in E_{down}} N_{e'}(t) \right) \times \frac{\eta}{2}, \quad t > 0$$

A general Hawkes LOB model – Signature plot



- ▶ Mean signature plot.
- ▶ Expected: signature plot of the Poisson model is flat
- ▶ 1-exponential and 2-exponential Hawkes: the realized volatility decreases when the sampling interval increases, but the long-term volatility level is too high compared to the data.

Inconsistencies between data and the linear Hawkes model

- Conditional probabilities of pairs $(X|Y)$ (probability of observing an event of type X given that the previous event is of type Y):

Pair	P_{simu}	P_{real}
$C_{buy}^1 L_{buy}^0$	0.402	0.048
$L_{buy}^1 L_{buy}^1$	1.628	0.141
$L_{sell}^1 L_{buy}^1$	1.288	0.171
$M_{sell}^0 C_{buy}^1$	0.545	0.068
$C_{buy}^1 C_{buy}^1$	0.548	0.072
$M_{sell}^1 C_{buy}^1$	0.854	0.037

- $C_{buy}^1 | L_{buy}^0$: needs two cancellations within the same microsecond ; $L_{buy}^1 | L_{buy}^1$ and $L_{sell}^1 | L_{buy}^1$: rare because spread constraints ; orders following a C_{buy}^1 : participants not willing to widen the spread.
- Further evidence : medians of L_1 norm of kernels $\phi \cdot C_{buy}^1$ in the 2-exponential model.

L_{buy}^0	L_{sell}^0	C_{buy}^0	C_{sell}^0	M_{buy}^0	M_{sell}^0
0.1563	0.2357	0.9392	0.0914	0	0
L_{buy}^1	L_{sell}^1	C_{buy}^1	C_{sell}^1	M_{buy}^1	M_{sell}^1
0.3845	0.1607	0	0	0.0013	0

Inhibition effects cannot be modelled in a framework with excitation only.

Inhibition in LOB models – Nonlinear Hawkes processes

- General non-linear Hawkes model:

$$\lambda(t) = \varphi \left(\lambda_0 + \int_0^t \Phi(t-s) dN_s \right).$$

- [15] modifies the previous model by including inhibition for the pairs identified above with a **negative 2-exponential kernels**:

$$\phi_{mn}(t) = \sum_{p=1}^2 -\alpha_{mnp} \exp(-\beta_{mnp}t),$$

where the α 's and β 's are non-negative real numbers, and the non-linear function $\varphi(x) = \max(x, 0)$ to keep the intensity positive.

- Some general results on non-linear Hawkes processes in e.g., [4, 24] (outside the scope of this course).

Nonlinear Hawkes processes – Conditional probabilities and inhibition kernels

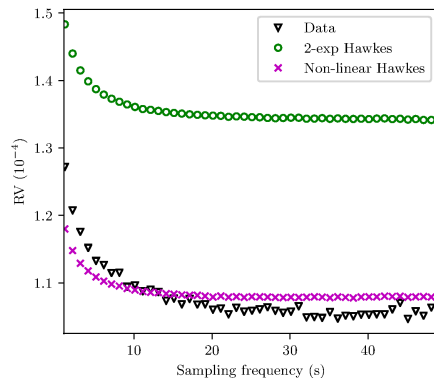
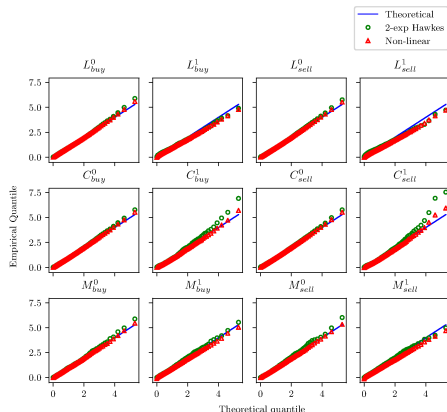
- Pairs previously identified as badly represented by the linear models:

Pair	P_{simu}	P_{simu}^{NL}	P_{real}
$C_{buy}^1 L_{buy}^0$	0.402	0.105	0.048
$L_{buy}^1 L_{buy}^1$	1.628	0.254	0.141
$L_{sell}^1 L_{buy}^1$	1.288	0.235	0.171
$M_{sell}^0 C_{buy}^1$	0.545	0.135	0.068
$C_{buy}^1 C_{buy}^1$	0.548	0.163	0.072
$M_{sell}^1 C_{buy}^1$	0.854	0.108	0.037

- Medians of L_1 norm of kernels $\phi.C_{buy}^1$ in non-linear model for those that were 0 in 2-exponential model:

M_{buy}^0	M_{sell}^0	C_{buy}^1	C_{sell}^1	M_{sell}^1
-0.0319	-0.1593	-0.0541	-0.1439	-0.1908

Nonlinear Hawkes processes – Goodness-of-fit and signature plots



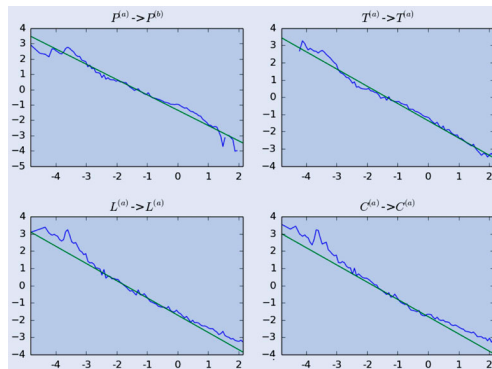
Inhibition effects exhibits satisfying goodness-of-fit and better reproduction of the volatility.

Non-parametric kernels – Estimation on futures data

- ▶ In [2], a Hawkes model with **non-parametric kernels** is fitted to the first limits with the following events $N_t = (P^{(a)}, P^{(b)}, T^{(a)}, T^{(b)}, L^{(a)}, L^{(b)}, C^{(a)}, C^{(b)})$ (mid-price moves, then trades, limit orders and cancellations that do not move the price)
- ▶ Data: Level-I data on BUND and DAX futures.
- ▶ Slowly decreasing kernels are found for many interactions (log-log-scale).

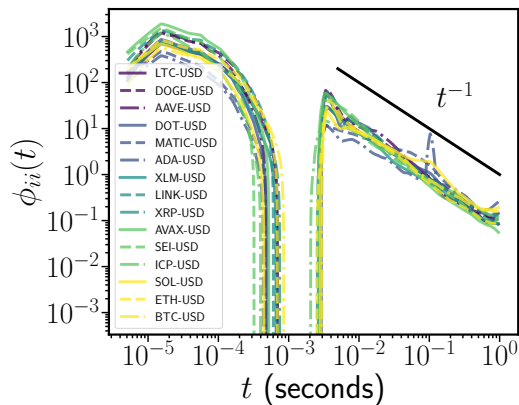
Exponential kernels are usually not found in empirical analysis of microstructure events.

- ▶ Multiple exponential kernels is a workaround for longer dependencies.



Source: [2]

Non-parametric kernels – Estimation on cryptocurrencies



- ▶ In [6] a 15-dimensional non-parametric Hawkes model is fitted to cryptocurrency trade data.
- ▶ Long memory observable on cryptocurrencies as well.
- ▶ Non-parametric estimation reveals complex structure of cross- and self-excitation in fitted Hawkes models.

Source : [6]

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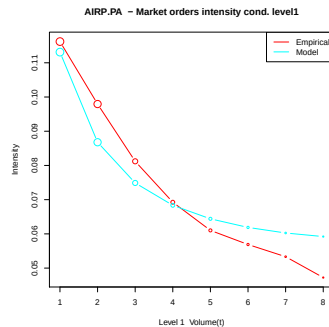
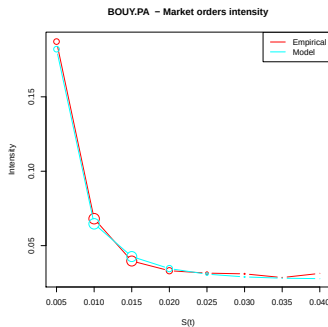
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State-dependent intensities

Empirical intensities of submission of market orders are state-dependent in e.g., spread (*left*) and the size of the best queue (*right*).



Source: [19]

- Light-blue curve : fit of a parametric model

$$\lambda^M(t; S(t), q_1(t)) = \exp \left[\beta_0 + \beta_1 \ln(S(t)) + \beta_{11} [\ln(S(t))]^2 + \beta_2 \ln(1 + q_1(t)) + \beta_{22} [\ln(1 + q_1(t))]^2 + \beta_{12} \ln(S(t)) \ln(1 + q_1(t)) \right].$$

A first state-dependent Hawkes models I

- ▶ [17] proposes a Hawkes extension in which the kernels depend on a state vector (e.g., representing the state of the LOB).
- ▶ Let $\mathcal{E}$ be a set of point processes with pooled event times $(T_n)_{n \in \mathbb{N}}$. Let E_n be the type of event observed at time T_n . The **intensity process** for the process $e \in \mathcal{E}$ is:

$$\lambda_e(t) = \nu_e + \sum_{e' \in \mathcal{E}} \int_{[0,t)} k_{ee'}(t-s, X_s) dN_{e'}(s)$$

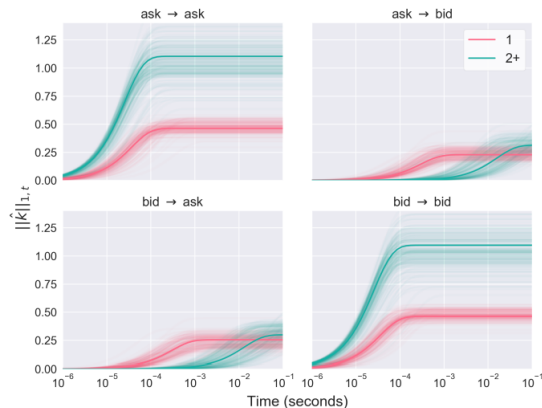
and the **state process** X is piecewise constant with jumps $(T_n)_{n \in \mathbb{N}}$ s.t.

$$\mathbf{P}(X(T_n) = x | E_n, \mathcal{F}_{T_n-}) = \Phi_{E_n}(X(T_n-), x),$$

where the Φ_{E_n} 's are transition probability matrices.

- ▶ Model is applied to imbalance/spread models for LOB events.

A first state-dependent Hawkes models II



(a) Models (state variable: bid-ask spread).

- ▶ Example of results for upward/downward events ($|\mathcal{E}| = 2$) and state=spread (*left*)
- ▶ State-dependency obvious.
- ▶ Potential parsimony problems.

Source : [17]

Another state-dependent Hawkes model I

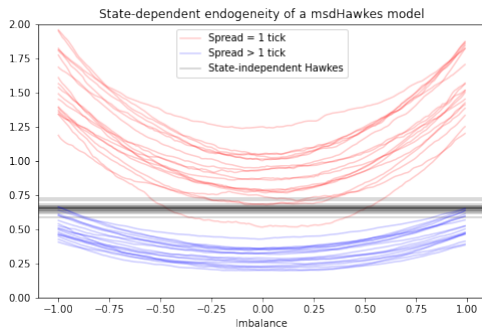
- ▶ [21] extends the standard Hawkes model with a state-dependent multiplicative factor. Let $d_e \in \mathbb{N}^*$. Let $(N_t)_{t \geq 0}$ be a d_e -dimensional counting process with stochastic intensity λ .
- ▶ **State representation** : Let $d_x \in \mathbb{N}$. Let $(X_t)_{t \geq 0}$ be an adapted, piecewise constant, left-continuous stochastic process, with values in $[-1, 1]^{d_x}$. $(X_t)_{t \geq 0}$ denotes the observable state space.
- ▶ **Hawkes process with multiplicative state-dependent factor (msdHawkes)** : point process N with an intensity $\lambda = (\lambda^1, \dots, \lambda^{d_e})$ of the form:

$$\lambda^e(t) = \left(\nu + \int_{]0, t[} \phi(t-s) \cdot dN_s \right) \exp(\langle \theta^e, X_t \rangle), \quad (2)$$

where $(\theta^1, \dots, \theta^{d_e}) \in (\mathbb{R}^{d_x})^{d_e}$ are the state coefficients.

- ▶ See also [23] for an analogous queue-size-dependent Hawkes order flow model.

Another state-dependent Hawkes model II



Importance of state-dependence, but potential complex interplay of regimes outside of standard theory.

A state-dependent spread Hawkes model I

- ▶ If $S_t^{+/-k}$ counts the number of positive/negative jumps of the spread of size k ticks and assuming the maximum jump size is K , the spread process is

$$S_t = S_0 + \sum_{k=1}^K k S_t^{+k} - \sum_{k=1}^K k S_t^{-k}.$$

- ▶ [20] propose a state-dependent spread Hawkes model:

$$\lambda_t^e = f^e(S_{t-}) \left[\mu^e + \sum_{e' \in \mathcal{E}} \int_0^t \Phi_{ee'}(t-s) dS_s^{e'} \right],$$

where $\mathcal{E} = \{-K, \dots, -1, 1, \dots, K\}$ is the space of jump sizes.

- ▶ The state-dependent function is one way to ensure that the spread remains positive: $f^{-k}(n) = 0$ if $n \leq k$.
- ▶ Normalization necessary (all quantities defined up to a constant because of f)

A state-dependent spread Hawkes model II

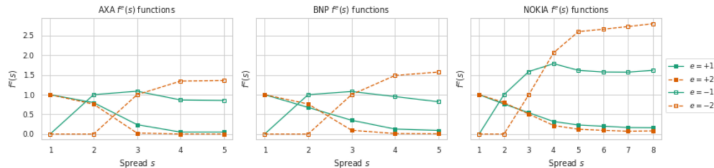


Figure 3: Estimations of the $\{f^e(s)\}_{e \in \mathcal{E}}$ functions for AXA (left), BNP (middle), NOKIA (right),

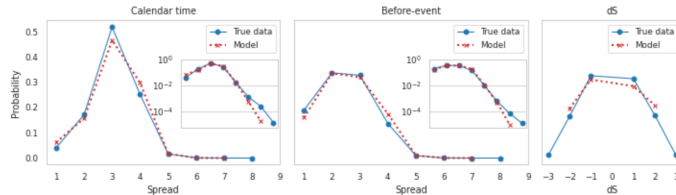


Figure 6: Spread distribution, comparison between true data and the data obtained through simulation for AXA. The left-hand figure is the Calendar time distributions, the middle figure is the event time distributions, and the right-hand figure is the distributions of spread jumps size.

- The state-dependent factor is crucial for accurate reproduction of empirical data.

Source : [20]

An intensity ratio model I

- ▶ One may also focus on the state-dependence by assuming a baseline intensity $\lambda_0(t)$, neither observable nor specified as a function of observables and parameters, common to all coordinates. Intensity $\lambda^i(t)$ of the process $(N_t^i)_{t \geq 0}$, $i \in \mathbb{I}$, is written.

$$\lambda^i(t, \vartheta) = \lambda_0(t) \lambda^i(X(t)),$$

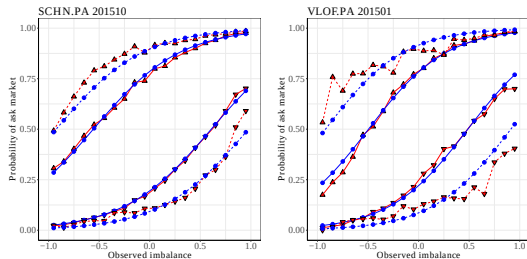
where $(X(t))_{t \geq 0}$ is an observable covariate process

- ▶ In the case $\lambda^i(X(t)) = \exp\left(\sum_{j \in \mathbb{J}} \vartheta_j^i X_j(t)\right)$ for a parameter vector $\vartheta = (\vartheta_j^i)_{i \in \mathbb{I}, j \in \mathbb{J}}$, MLE estimation of the *relative* dependencies available via the **intensities ratios** ([18]):

$$r_i(t, \theta) = \frac{\lambda^i(t, \vartheta)}{\sum_{i' \in \mathbb{I}} \lambda^{i'}(t, \vartheta)} = \left[1 + \sum_{i' \in \mathbb{I} \setminus \{i\}} \exp\left(\sum_{j \in \mathbb{J}} (\vartheta_j^{i'} - \vartheta_j^i) X_j(t)\right) \right]^{-1}.$$

- ▶ $\lambda^i(X(t))$ can also be flexibly represented with a neural network.

An intensity ratio model II



Examples of fitted order flows probabilities in a ratio model.

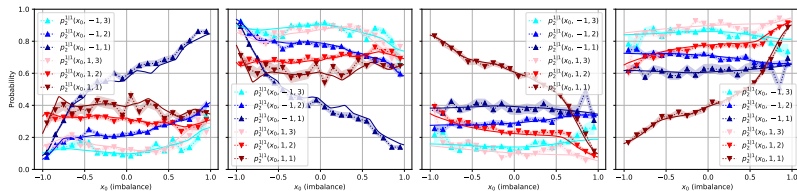


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Lab 4

Lab 4 - Questions

Remember to keep the data file structure `./Data/stocki_dayj_trades.gz` as asked in Lab 1.

1. **Brownian motions with Poisson sampling.** Take your results from Lab 2 Question 2 and add the Hayashi-Yoshida estimator to the plot. Comment.
2. **Signature plots and Epps effect.** On your dataset, estimate the integrated realized variances for each stock and the correlation between each pair of stocks using the standard estimators and the Hayashi-Yoshida estimators. Comment.
3. **Lead-lag estimation.** On your dataset, for each pair of stock, estimate the lead-lag relationship using the Hayashi-Yoshida estimators. Comment.
4. **A Hawkes process for trades.** Is a Hawkes process a good model for the time dynamics of the trades in your dataset ? Explain your experiments *in details* and use statistical arguments to support your answers.
5. **Hawkes fit on signature plots.** Fit the single stock Hawkes model of [1] on the signature plot for each stock of your dataset. Comment.

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